We create chemistry

# Safety data sheet



## 1. Identification

### Product identifier

## Catalyst H0-35 / H0-31 regenerated

### Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: catalyst
Recommended use: Industrial catalyst

### Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Petrochemicals

Telephone: +49 621 60-42151
E-mail address: sds-petrochemicals@basf.com

### Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

## 2. Hazards Identification

### Classification of the substance or mixture

According to UN GHS criteria

Aquatic Acute 2
Aquatic Chronic 2

For the classifications not written out in full in this section the full text can be found in section 16.

### Label elements

Globally Harmonized System (GHS)

Pictogram:

Hazard Statement:

| | |
|---|---|
| H401 | Toxic to aquatic life. |
| H411 | Toxic to aquatic life with long lasting effects. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P273 | Avoid release to the environment. |

Precautionary Statements (Response):

| | |
|---|---|
| P391 | Collect spillage. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

Labeling of special preparations (GHS):
The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 0 - 1 %, Inhalation - dust

### Other hazards

According to UN GHS criteria

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

See section 12 - Results of PBT and vPvB assessment.

## 3. Composition/Information on Ingredients

### Substances

Not applicable

### Mixtures

Chemical nature

catalyst

Preparation based on: metal oxides, on inorganic substrate

Hazardous ingredients (GHS)
According to UN GHS criteria

Disilver oxide

| | |
|---|---|
| Content (W/W): >= 0 % - <= 0,1 % | Ox. Sol. 1 |
| CAS Number: 20667-12-3 | Acute Tox. 5 (oral) |
| EC-Number: 243-957-1 | Eye Dam./Irrit. 1 |
| | Aquatic Acute 1 |
| | Aquatic Chronic 1 |
| | M-factor acute: 100 |
| | M-factor chronic: 100 |
| | H271, H318, H303, H400, H410 |

For the classifications not written out in full in this section the full text can be found in section 16.

## 4. First-Aid Measures

### Description of first aid measures

Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Rinse mouth and then drink 200-300 ml of water.

### Most important symptoms and effects, both acute and delayed

Symptoms: (Further) symptoms and / or effects are not known so far

Hazards: (Further) symptoms and / or effects are not known so far

### Indication of any immediate medical attention and special treatment needed

Treatment: Symptomatic treatment (decontamination, vital functions).

## 5. Fire-Fighting Measures

### Extinguishing media

Suitable extinguishing media:
foam, dry powder, water spray

Unsuitable extinguishing media for safety reasons:
water jet, carbon dioxide

Additional information:
Use extinguishing measures to suit surroundings.

If a metal fire occurs, use dry sand, dry powder for Class D or cement.

### Special hazards arising from the substance or mixture

Cool endangered containers with water-spray. See SDS section 7 - Handling and storage.

### Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus. Special protective equipment for firefighters

Further information:
Evacuate area of all unnecessary personnel. Fight fire from maximum distance.

Extend fire extinguishing measures to the surroundings.

## 6. Accidental Release Measures

Avoid dispersal of dust in the air (e.g. by clearing dusty surfaces with compressed air). Dust in sufficient concentration can result in an explosive mixture in air. Handle to minimize dusting and eliminate open flame and other sources of ignition.

### Personal precautions, protective equipment and emergency procedures

Avoid dust formation. Do not breathe dust. Avoid contact with the skin, eyes and clothing. Use personal protective clothing. Information regarding personal protective measures, see section 8.

### Environmental precautions

Do not discharge into drains/surface waters/groundwater. Contain contaminated water/firefighting water.

### Methods and material for containment and cleaning up

Avoid raising dust. Dampen, pick up mechanically and dispose of. Dispose of absorbed material in accordance with regulations. Reclaim for processing if possible.

## 7. Handling and Storage

### Precautions for safe handling

Handle in accordance with good industrial hygiene and safety practice. Avoid the formation and build up of dust - danger of dust explosion.

Protection against fire and explosion:
No special precautions necessary.

### Conditions for safe storage, including any incompatibilities

No applicable information available.
Further information on storage conditions: Containers should be stored tightly sealed in a dry place.

### Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

## 8. Exposure Controls/Personal Protection

### Control parameters

Components with occupational exposure limits

1344-28-1: Aluminium oxide
20667-12-3: Disilver oxide

### Exposure controls

Personal protective equipment

Respiratory protection:
Breathing protection if dusts are formed. Particle filter with low efficiency for solid particles (e.g. EN 143 or 149, Type P1or FFP1)

Hand protection:
Chemical resistant protective gloves (EN ISO 374-1)

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
Ensure adequate ventilation. Avoid inhalation of dusts. Avoid contact with the skin, eyes and clothing. Wearing of closed work clothing is required additionally to the stated personal protection equipment. Handle in accordance with good industrial hygiene and safety practice. At the end of the shift the skin should be cleaned and skin-care agents applied.

## 9. Physical and Chemical Properties

### Information on basic physical and chemical properties

| | |
|---|---|
| Form: | extrudates |
| Colour: | grey<br>brown |
| Odour: | odourless |
| Odour threshold: | not applicable, odour not perceivable |
| pH value: | approx. 7 - 8<br>(20 °C) |
| Melting point: | not determined |

| | | |
|---|---|---|
| Boiling point: | not relevant | |
| Flash point: | not applicable, the product is a solid | |
| Evaporation rate: | The product is a non-volatile solid. | |
| Flammability: | not flammable | (derived from flash - and boiling point) |
| Lower explosion limit: | For solids not relevant for classification and labelling. | |
| Upper explosion limit: | For solids not relevant for classification and labelling. | |
| Ignition temperature: | not relevant | |
| Vapour pressure: | not applicable | |
| Density: | No data available. | |
| Relative density: | No data available. | |
| Relative vapour density (air): | The product is a non-volatile solid. | |
| Solubility in water: | not soluble, immiscible | |
| Partitioning coefficient n-octanol/water (log Kow): | The value has not been determined because the substance is inorganic. | |
| Self ignition: | not self-igniting | |
| Thermal decomposition: | No decomposition if correctly stored and handled. | |
| Viscosity, dynamic: | not applicable, the product is a solid | |
| Viscosity, kinematic: | not applicable, the product is a solid | |
| Explosion hazard: | not explosive | |
| Fire promoting properties: | not fire-propagating | |

### Other information

| | | |
|---|---|---|
| Self heating ability: | It is not a substance capable of spontaneous heating. | |
| Radioactivity: | | not radioactive for transport purposes |
| Bulk density: | approx. 650 - 750 kg/m3 | |
| Hygroscopy: | Non-hygroscopic | |
| Grain size distribution: | No data available. | |

## 10. Stability and Reactivity

### Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

| | | |
|---|---|---|
| Corrosion to metals: | No corrosive effect on metal. | |
| Reactions with water/air: | Reaction with: | water |
| | Flammable gases: | no |
| | Toxic gases: | no |
| Formation of flammable gases: | Remarks: | Forms no flammable gases in the presence of water. |

### Chemical stability
The product is stable if stored and handled as prescribed/indicated.

Peroxides: The product does not contain peroxides.

### Possibility of hazardous reactions
No hazardous reactions if stored and handled as prescribed/indicated.

### Conditions to avoid
Avoid dust formation.

### Incompatible materials

Substances to avoid:
reducing agents

### Hazardous decomposition products

Possible decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

## 11. Toxicological Information

### Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Virtually nontoxic after a single ingestion. Virtually nontoxic after a single skin contact. Virtually nontoxic by inhalation.

The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 0 - 1 %

Irritation

Assessment of irritating effects:
Not irritating to eyes and skin.

Respiratory/Skin sensitization

Assessment of sensitization:
No sensitizing effect.

Germ cell mutagenicity

Assessment of mutagenicity:
Mutagenicity tests revealed no genotoxic potential.

Carcinogenicity

Assessment of carcinogenicity:
The whole of the information assessable provides no indication of a carcinogenic effect.

Reproductive toxicity

Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:
No indications of a developmental toxic / teratogenic effect were seen in animal studies.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
No adverse effects were observed after repeated exposure in animal studies.

Aspiration hazard

No aspiration hazard expected.

Other relevant toxicity information

The product has not been tested. The statement has been derived from the properties of the individual components.

## 12. Ecological Information

### Toxicity

Assessment of aquatic toxicity:
Acutely toxic for aquatic organisms. Toxic to aquatic organisms based on long-term (chronic) toxicity study data.

*Information on: Disilver oxide*
*Assessment of aquatic toxicity:*

*Very toxic (acute effect) to aquatic organisms. Very toxic to aquatic organisms based on long-term (chronic) toxicity study data.*
*The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
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*Information on: Disilver oxide*
*Toxicity to fish:*
*LC50 (96 h) 0,0012 mg/l, Pimephales promelas (Fish test acute, semistatic)*
*The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
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*Information on: Disilver oxide*
*Aquatic invertebrates:*
*LC50 (48 h) 0,00022 mg/l, Daphnia magna (Daphnia test acute, semistatic)*
*The statement of the toxic effect relates to the analytically determined concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
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*Information on: Disilver oxide*
*Aquatic plants:*
*EC10 (24 h) 0,00041 mg/l (growth rate), Pseudokirchneriella subcapitata (other, Flow through.)*
*The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
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*Information on: Disilver oxide*
*Microorganisms/Effect on activated sludge:*
*No observed effect concentration (13,3 min) 0,025 mg Ag/L, nitrifying bacteria (other, static)*
*The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
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*Information on: Disilver oxide*
*Chronic toxicity to fish:*
*EC10 (196 d) 0,17 µg Ag/L, Oncorhynchus mykiss (OECD Guideline 210, Flow through.)*
*The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
---

*Information on: Disilver oxide*
*Chronic toxicity to aquatic invertebrates:*
*EC10 (21 d) 0,00214 mg/l, Daphnia magna (other, semistatic)*
*The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
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## Persistence and degradability

Assessment biodegradation and elimination (H2O):
Not applicable for inorganic substances.

## Bioaccumulative potential

*Information on: Disilver oxide*
*Assessment bioaccumulation potential:*
*Does not significantly accumulate in organisms.*
*The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
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*Information on: Disilver oxide*
*Bioaccumulation potential:*
*Bioconcentration factor: approx. 70 (41 d), Cyprinus carpio (measured)*
*The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
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### Mobility in soil

*Information on: Disilver oxide*
*Assessment transport between environmental compartments:*
*Volatility: No data available.*
*Adsorption in soil: Adsorption to solid soil phase is expected.The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
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### Results of PBT and vPvB assessment

PBT assessment does not apply. Not applicable for inorganic substances.

### Other adverse effects

The product does not contain substances that are listed in Annex I of Regulation (EC) 2037/2000 on substances that deplete the ozone layer.

### Additional information

Other ecotoxicological advice:
The product has not been tested. The statement has been derived from the properties of the individual components. The product has been assessed on the basis of the components' available data. To some extent data gaps exist for individual components. According to our present knowledge and experience dangers which are not covered by the current labeling are not to be expected.

## 13. Disposal Considerations

### Waste treatment methods

Dispose of in accordance with national, state and local regulations.
Contact specialized companies about recycling.

Contaminated packaging:
Disposal must be made according to official regulations.

## 14. Transport Information

**Land transport**

ADR

| | |
|---|---|
| UN number or ID number: | UN3077 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (DISILVER OXIDE) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

RID

| | |
|---|---|
| UN number or ID number: | UN3077 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (DISILVER OXIDE) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

**Inland waterway transport**

ADN

| | |
|---|---|
| UN number or ID number: | UN3077 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (DISILVER OXIDE) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

Transport in inland waterway vessel
Not evaluated

**Sea transport**

IMDG

| | |
|---|---|
| UN number or ID number: | UN 3077 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (DISILVER OXIDE) |

| | |
|---|---|
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes<br>Marine pollutant: YES |
| Special precautions for user: | EmS: F-A; S-F |

**Air transport**

IATA/ICAO

| | |
|---|---|
| UN number or ID number: | UN 3077 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (DISILVER OXIDE) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

### Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

## 15. Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

Not applicable

## 16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

| | |
|---|---|
| Aquatic Acute | Hazardous to the aquatic environment - acute |
| Aquatic Chronic | Hazardous to the aquatic environment - chronic |
| Ox. Sol. | Oxidising solids |
| Acute Tox. | Acute toxicity |
| Eye Dam./Irrit. | Serious eye damage/eye irritation |
| H271 | May cause fire or explosion; strong oxidizer. |
| H318 | Causes serious eye damage. |
| H303 | May be harmful if swallowed. |
| H400 | Very toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the

responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.